ASSIGNMENT COVER SHEET

Course Code and Description	UECS1413 DATABASE SYSTEM FUNDAMENTALS
Lecturer Name	Dr.Sugumaran a/l Nallusamy
Assignment Title	Assignment 1
DECLARATION	
We declare that this is a group assignment and that no part of this submission has been copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part been written for us by another person.	

Programme	Student ID Numbers	Student Names	Practical Group
Software Engineering	2301521	Ang Yong Xen	P5
Software Engineering	2302120	Kurt Lai Ee Hern	P9
Software Engineering	2302452	Kyle Yamato Christian	P5
Software Engineering	2302533	Tan Jun Yin	P5

Description	Max Marks	Marks Obtained
Correct requirement identified from the case study given	30	
Reasonable Assumptions	10	
Identifying Entities with a proper unique identifier	20	
Identifying the Relationship, Generalisation/Specialisation hierarchies	20	
Multiplicity	20	
Total	100	

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System Requirement

Customer

1. The customer must be able to create an account in the system, where their details, including Customer_ID (Primary Key), customer name, email, contact number, and address will be stored.
2. The customers can be categorized into three types, regular customers, agents, and drop shippers. Agents and drop shippers will have their names and contact information recorded separately.
3. The agent and drop shipper will get a 3% commission for Biodegradable Packaging products and a 10% commission for Sharifah Ready to Eat products calculated by the system.
4. The customer must be able to place orders to purchase one or more products from Sharifah Ready to Eat products and Biodegradable Packaging products.
5. The customer must be able to place orders through WhatsApp and e-commerce platforms such as Shopee and Lazada.
6. The customer must be able to select from multiple delivery options, including HIT's in-house delivery and third-party delivery services such as Grab and Lalamove.
7. The customer must be able to receive one or more invoices after successfully making a payment through cash on delivery, online banking, credit or debit cards, or an Ewallet.

Product

1. The product can be considered a superclass, at the same time this superclass contains two subclasses, which are the Sharifah Ready to Eat products and Biodegradable Packaging products.
2. The products must be categorized into Sharifah Ready to Eat products and Biodegradable Packaging products, each product must be identified by a unique Product_ID (Primary Key).
3. Each Sharifah Ready to Eat product must have attributes including product name, weight, price, cartons, and quantity.
4. The Sharifah Ready to Eat products will be categorized into three categories, such as ready to eat, ready to cook, and paste.

5. Each Biodegradable Packaging product must have attributes including item descriptions, cartons, packets, and price.
6. The Biodegradable Packaging products will be categorized into four categories, such as cups, containers, utensils, and carry bags.
7. The system must allow authorized users to add, update, and remove product details as required,
8. The system must automatically update inventory levels based on sales transactions and restocking activities.
9. The vendors will supply the Sharifah Ready to Eat products and Biodegradable Packaging products.
10. The Sharifah Ready to Eat products and Biodegradable Packaging products will be stored in the inventory.

Orders

1. Each order has its own unique Order_ID (Primary Key), including order date, order commission rate and order discount.
2. Each order must contain at least one or more order details.
3. Each order detail must refer to one or more products.
4. The employee must process orders by assigning a unique Employee_ID (Primary Key) to each order, ensuring that multiple products with different quantities can be included in a single order.
5. Each order must require one delivery, and each delivery must require only one order.
6. The system must record all payments and validate that payments are only processed through HIT's company account.

Delivery

1. Each delivery has its own unique Delivery_ID (Primary Key), including delivery fees and methods.
2. Each delivery must be linked to an order.

3. The customer must be able to select their preferred delivery methods, either through HIT’s in-house delivery services (available in specific regions) or third-party services such as Grab and Lalamove.
4. The system must dynamically calculate delivery fees and include them in the final amount of order cost.

Vendor

1. The system must store vendor details, including VendorID (Primary Key), vendor name, contact information, bank details, vendor category (Sharifah Food Sdn Bhd / Other), and track product supplies and their respective minimum order quantities.
2. The HIT SDN BHD will order supplies from the vendors which are Sharifah Food Sdn Bhd for Sharifah Ready to Eat products and multiple vendors of Biodegradable Packaging companies or stores for Biodegradable Packaging products.
3. The system must record and track vendor payments, restricting transactions that are available through bank transfers only.

Inventory

1. Each inventory record is uniquely identified by Inventory_ID (Primary Key), including Inventory category, Stock level and Restock date.
2. The system will store the product stock levels of the Sharifah Ready to Eat products and Biodegradable Packaging products.
3. The system must monitor inventory levels, generate stock alerts for low inventory, and provide recommendations for reordering the products.
4. The system must allow authorized users to update inventory records when new stock arrives and when the products are discontinued.
5. The system must generate inventory reports showing the current stock levels, stock movement history, and predicted restocking needs.

Employee

1. The employee must handle at least one or more orders.
2. Each employee is uniquely identified by Employee_ID (Primary Key), including Employee name and contact.
3. The employees are responsible for overseeing order fulfilment, including coordinating deliveries.

Additional System Requirements

1. The system must maintain an audit log to track all activities, including user actions, transactions, and data modifications, ensuring accountability.
2. The system must support multiple languages such as English, Bahasa Melayu, and Chinese for both customers and employees to choose from.
3. The system must allow the customers to contact customer services directly through the platform, with selected options such as live chat, email support, or phone calls.

Assumptions

1. Assume that the target customers for Sharifah Ready to Eat products are travelers and for Biodegradable Packaging products are hotels.
2. Assume inventory levels will be automatically updated when an order is placed, ensuring real-time stock tracking.
3. Assume that if a customer cancels an order, the system must immediately update the order status to prevent invoice generation.
4. Assume that there are promotions involved for both agent and drop shipper, the price of the products will be changed from time to time.
5. Assume that all customer data is accurate without any duplicate records to ensure the validity of each customer's information.
6. Assume that automated email notifications must be generated for key events, including order placement, invoice generation, and delivery status updates.
7. Assume the user role permissions must be strictly enforced, with the system having full control, the employees will handle the orders.

Entity Relationship Modelling (ER Diagram)